Landau Theory of Phase Transitions

1 Introduction

Landau theory is a phenomenological framework developed by Lev Landau to describe continuous (second-order) phase transitions. Rather than focusing on microscopic details, the theory is based on symmetry arguments and the concept of an *order parameter* that characterizes different phases of matter.

Landau theory successfully explains qualitative features of phase transitions, such as spontaneous symmetry breaking, critical behavior, and universality, though it fails to accurately predict critical exponents in low-dimensional systems.

2 Order Parameter

An *order parameter* is a quantity that:

- Is zero in the disordered (high-symmetry) phase
- Is nonzero in the ordered (low-symmetry) phase
- Changes continuously at a second-order phase transition

Examples:

- Ferromagnetism: magnetization M
- Liquid–gas transition: density difference $\rho - \rho_c$
- Superconductivity: complex order parameter ψ

3 Landau Free Energy Expansion

The central assumption of Landau theory is that near the critical temperature T_c, the free energy can be expressed as an analytic expansion in the order parameter.

For a scalar order parameter ϕ, the Landau free energy density is written as:

$$F(\phi) = F_0 + a\phi^2 + b\phi^4 + c\phi^6 + \cdots \tag{1}$$

where:

- F_0 is the free energy of the disordered phase
- a, b, c are phenomenological coefficients
- Stability requires $b > 0$ (or $c > 0$ if $b = 0$)

Odd powers of ϕ are excluded when the system has $\phi \to -\phi$ symmetry.

4 Temperature Dependence

The coefficient a is assumed to vary linearly with temperature:

$$a(T) = a_0(T - T_c) \tag{2}$$

where $a_0 > 0$. Thus:

- $a > 0$ for $T > T_c$
- $a < 0$ for $T < T_c$

This change in sign drives the phase transition.

5 Minimization of Free Energy

The equilibrium value of the order parameter is obtained by minimizing the free energy:

$$\frac{dF}{d\phi} = 0 \tag{3}$$

Using

$$F(\phi) = a\phi^2 + b\phi^4 \tag{4}$$

we find:

$$\frac{dF}{d\phi} = 2a\phi + 4b\phi^3 = 0 \tag{5}$$

The solutions are:

$$\phi = 0 \quad \text{or} \quad \phi^2 = -\frac{a}{2b} \tag{6}$$

5.1 High-Temperature Phase ($T > T_c$)

Here $a > 0$, so the only stable solution is:

$$\phi = 0 \tag{7}$$

This corresponds to the symmetric, disordered phase.

5.2 Low-Temperature Phase ($T < T_c$)

Here $a < 0$, and the free energy is minimized by:

$$\phi = \pm\sqrt{-\frac{a}{2b}} \tag{8}$$

This represents spontaneous symmetry breaking.

6 Critical Behavior

Near T_c, the order parameter behaves as:

$$\phi \propto (T_c - T)^{1/2} \tag{9}$$

This defines the critical exponent:

$$\beta = \frac{1}{2} \tag{10}$$

Other critical exponents predicted by Landau theory include:

- Heat capacity: $\alpha = 0$
- Susceptibility: $\gamma = 1$
- Field dependence: $\delta = 3$

These are *mean-field* values.

7 First-Order Phase Transitions

If $b < 0$, the ϕ^4 theory becomes unstable and higher-order terms must be included:

$$F(\phi) = a\phi^2 + b\phi^4 + c\phi^6 \tag{11}$$

In this case:

- The order parameter changes discontinuously
- Latent heat is present
- The transition is first-order

8 Effect of an External Field

Including an external field h conjugate to ϕ:

$$F(\phi) = a\phi^2 + b\phi^4 - h\phi \tag{12}$$

This removes the degeneracy between $\phi > 0$ and $\phi < 0$ and smooths out the phase transition.

9 Landau-Ginzburg Theory

Landau theory can be generalized to spatially varying order parameters:

$$F[\phi] = \int d^dx \left[a\phi^2 + b\phi^4 + \kappa(\nabla\phi)^2\right] \tag{13}$$

This formulation allows the study of:

- Domain walls
- Correlation lengths
- Critical fluctuations

10 Limitations of Landau Theory

- Neglects thermal and quantum fluctuations
- Fails in low dimensions ($d \leq 2$)
- Incorrect critical exponents near T_c

These issues are resolved by the renormalization group theory.

11 Conclusion

Landau theory provides a powerful and elegant framework for understanding phase transitions through symmetry and order parameters. While approximate, it forms the conceptual foundation for modern theories of critical phenomena and spontaneous symmetry breaking.